ELEC-337
ELECTRONICS LABORATORY-II

EXPERIMENT 2

Single-BJT Amplifiers at Low and High Frequencies

Name:.....

Surname: .....

Student ID:.....

Section:.....

Date:.....

OBJECTIVE: *Exploring the behavior of capacitor-coupled BJT amplifiers at low-frequencies, and examining the high-frequency behavior of the BJT itself in a simple circuit.*

DEPARTMENT OF ELECTRONICS ENGINEERING
GEBZE TECHNICAL UNIVERSITY

This experiment has been adopted from Department of Electrical and Electronics Engineering, Boğaziçi University

PROCEDURE

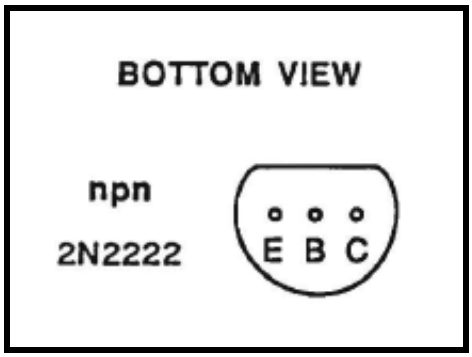


Figure 1. 2N2222 BJT Pin Configuration

1.The Basic Common-Emitter (CE) Circuit

1.1 The DC Situation:

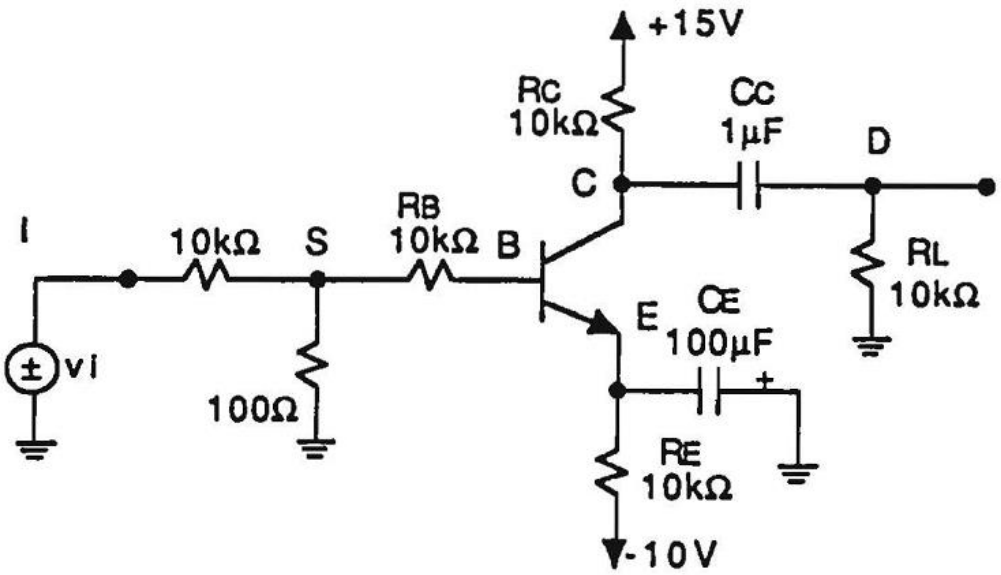


Figure 2. A Basic CE BJT Amplifier.

Assemble the circuit shown in Figure 2. Note the resistance values. With input (node I) grounded, measure the DC voltages at nodes S, B, E and C. Use these values to calculate bias currents, ß, α, r_e, r_π and g_m. We will assume V_A to be very large, as it normally is, and ignore r_o as a consequence.

Table 1. DC values for CE BJT amplifier circuit.

V_S	V_B	V_E	V_C	I_E	I_C
…….	…….	…….	…….	…….	…….
I_B	β	α	r_e	r_π	g_m
…….	…….	…….	…….	…….	…….

1.2 Mid-Band Response:

For the circuit in Figure 2, apply a sine-wave signal of 1 Vpp at 1 kHz to node I. Measure peak-to-peak voltages at nodes S, B, E, C, D using your oscilloscope.

Table 2. Peak-to-peak values for CE BJT amplifier circuit.

υ_s	υ_b	υ_e	υ_c	υ_d
…….	…….	…….	…….	…….

2.Low Frequency Response

2.1 Basic Overall Response:

Use the same circuit as E.1.2 (Figure 2), with a sine-wave signal of 1 Vpp at 1 kHz at node I. With probes connected to nodes S and D, lower the frequency slowly and note the corresponding peak-to-peak voltages. After completing the measurements, find the lower 3 dB frequency, f_L.

Table 3. Peak-to-peak voltages for corresponding frequencies.

	υ_s	υ_d	υ_d/υ_s
1 kHz	…….	…….	…….
500 Hz	…….	…….	…….
200 Hz	…….	…….	…….
100 Hz	…….	…….	…….
50 Hz	…….	…….	…….
20 Hz	…….	…….	…….
10 Hz	…….	…….	…….
5 Hz	…….	…….	…….

f_L	…….

3.High Frequency Operation of the Common-Emitter (CE) Amplifier

3.1 Basic AC Measurements:

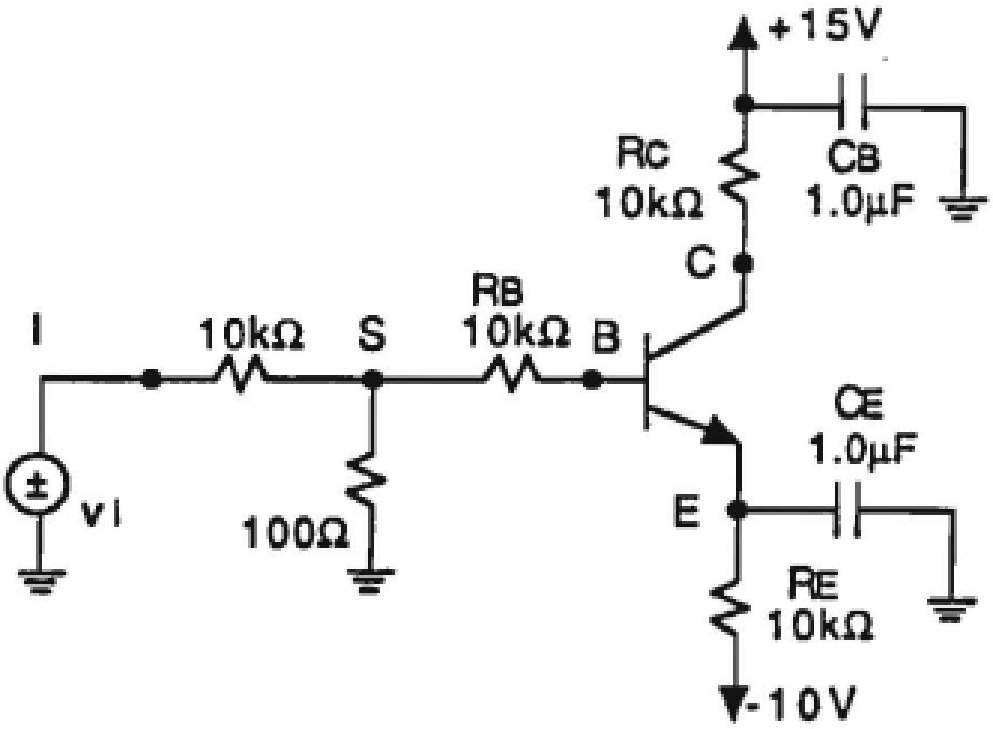


Figure 3. Modifications for high-frequency operation.

Assemble the circuit in Figure 3 with a 4 Vpp sine wave input at 10 kHz. As the frequency is varied, ensure that the input signal amplitude (at node I) holds constant. Adjust the generator output as necessary.

a) Using the oscilloscope at AC coupling, measure the peak-to-peak voltages at nodes S, B and C.
b) Now, noting the signal at node C in particular, raise the frequency, until the voltage at C falls by 3 dB, to 0.707 of its mid-band value. Note the frequency as f_1.
c) Raise the frequency to $10f_1$, and measure the peak-to-peak voltages at nodes S, B and C.
d) Now, shunt R_C by a second 10 kΩ resistor, and proceed to find the modified 3 dB frequency, f_2.

Table 4. Peak-to-peak values and gains for two cases.

R_C	Input	f	v_s	v_b	v_c
10 kΩ	10 kHz	10 kHz	……	……	……
	f_1	……	……	……	……
	$10f_1$	……	……	……	……
5 kΩ	10 kHz	10 kHz	……	……	……
	f_2	……	……	……	……
	$10f_2$	……	……	……	……

REFERENCES

- Smith, K. C. and Sedra, A. S., *Laboratory Explorations for Microelectronic Cirtuits*, Oxford University Press, 1998.

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